Narrative Twist (NT) — method & output columns

What the algorithm computes

For a **target sentence** S_2 in a story, we ask how much it is a real narrative twist: does S_2 make the continuation S_3 far more likely than a *banal* sentence would? We compare the real continuation probability (*posterior*) to the average over model-generated banal alternatives (*prior*):

$$\text{nt} = \log \frac{P(S_3 \mid S_1, S_2)}{Q}, \qquad Q = \frac{1}{n} \sum_{i=1}^{n} P\big(S_3 \mid S_1, S_2^{*i}\big) .$$

A high `nt` means S_2 is a strong twist. Sentence probabilities are **per token** (length-normalised, perplexity-style): $\log P(S_3 \mid \cdot) = \frac{1}{m} \sum_{j=1}^{m} \log P(\text{token}_j \mid \text{tokens}_{<j})$.

split the story into sentences $S_1, \ldots, S_k$
for each target S_t with $t = 2, \ldots, k-1$ **do** ▷ needs a sentence before and after
    $S_1 \leftarrow S_1 \cdots S_{t-1}$ ▷ context = all preceding sentences
    generate n=10 banal alternatives $S_2^{*1}, \ldots, S_2^{*n}$ of S_t ▷ from S_1, temperature 1
    for $S_3 \in \{\,\textsc{next} = S_{t+1},\ \ \textsc{all} = S_{t+1} \cdots S_k\,\}$ **do**
        $P \leftarrow P(S_3 \mid S_1, S_t)$ ▷ posterior, per token
        $Q \leftarrow \frac{1}{n} \sum_i P(S_3 \mid S_1, S_2^{*i})$ ▷ prior, per token
        nt $\leftarrow \log(P/Q)$ ▷ the score
    end for
end for

Outputs. The **main file** `<name>.xlsx` has one row per sentence and the score under each continuation mode:

Column	Meaning
`nt_next` (NT_A)	score with $S_3 = S_{t+1}$ (next sentence only): $\log\big(P(S_{t+1} \mid S_1, S_2)/Q\big)$.
`nt_all` (NT_B)	score with $S_3 = S_{t+1} \cdots S_k$ (all following sentences): $\log\big(P(S_{t+1\cdots k} \mid S_1, S_2)/Q\big)$.

The **debug file** `<name>_debug.xlsx` expands this (one row per sentence *and* mode), with the intermediate values — columns below.

Debug file columns

Column	Meaning	Formula
participant	Participant ID.	—
story	Which story (story, or text_1/text_2).	—
sent_idx	Position of S_2 (1 = 2nd sentence; first/last never scored).	—
sentence	Text of the target sentence S_2.	—
s3_mode	Continuation: next (S_{t+1}) or all ($S_{t+1} \cdots S_k$).	—
nt	The score.	$\log(P/Q) = \texttt{log\_post} - \texttt{log\_prior}$
log_post	Per-token log-prob of S_3 with the real S_2.	$\log P(S_3 \mid S_1, S_2)$
log_prior	Log of the mean prior over alternatives.	$\log Q$
ratio	How many times more likely S_3 is.	$P/Q = e^{\texttt{log\_post}-\texttt{log\_prior}}$
num_alternatives	Number of banal alternatives S_2^* used to estimate Q.	—

Alternatives file (validity check)

<name>_alternatives.xlsx lists, for every sentence and mode, the real sentence and each generated banal alternative S_2^*, with its own continuation log-probability — so one can check that the real sentence beats the banal ones.

Column	Meaning
role	real_S2 (the actual sentence) or alternative (a generated S_2^*).
alt_id	0 for the real sentence; $1, 2, \ldots$ for the alternatives.
sentence	Text of the real sentence or of the alternative.
log_prob	Per-token $\log P(S_3 \mid S_1, \cdot)$: log_post for the real, the prior for an alternative.

How the columns relate (a self-check, true on every row):

- the score is just the difference of the two log-probabilities: $\texttt{nt} = \texttt{log\_post} - \texttt{log\_prior}$ (because $\log P - \log Q = \log(P/Q)$);
- ratio is the same thing on the ordinary (non-log) scale: $\texttt{ratio} = e^{\texttt{nt}} = P/Q$ (how many times more likely S_3 is with the real S_2).